PLANNING DOCUMENT

GENERAL HOSPITAL BKD

PHARMACY SYSTEM Architecture & Page Design Plan

A practical blueprint for the first release, pharmacy workflows, local and cloud operation, and the user interface concept.

PLANNING SCOPE

- Drug catalogue, stock and inter-unit transfers
- POS sales and payment recording
- Patient prescriptions and dispensing
- Physical stock counts and reconciliation
- Offline operation, staff accounts and text chat

Prepared for General Hospital BKD | Kowaguru Technology Limited

18 October 2026 | Version 1.0 | Requirements and designs for review

Confirmed scope & module map

CLIENT CONFIRMATIONS

Initial feature scope accepted
Allow more pharmacy units and staff
Hospital server, with cloud support
MD remote access through VPN
Fixed medicine prices
NHIS can follow in a later release
Existing stock and price records exist
Hardware selection: recommend suitable devices

FIRST RELEASE MODULES

Catalogue and fixed-price setup
Receiving, balances and transfers
POS, receipts and payment methods
Patient records and prescriptions
Physical stock count and adjustment
Low-stock and near-expiry alerts
Staff accounts, roles and reports
Offline sales queue and synchronization
Text-only staff-to-staff chat

LATER PHASE

DEFERRED / OPTIONAL MODULES

NHIS claims; controlled-drug registers; broader hospital integrations

Advanced management reports; formal receiving approvals; POS terminal integration

Physical count is important to the client and is included in the first release plan.

HARDWARE STARTING RECOMMENDATION

PERIPHERALS TO CONFIRM WITH THE HOSPITAL

USB 1D/2D barcode scanner for medicine packs and product codes

Thermal receipt printer (for example, Epson TM-T20 series or equivalent)

Direct thermal label printer for patient medicine labels

UPS for the local server, network equipment and pharmacy workstations

Confirm Windows, paper width, drivers, barcode format and service availability before purchasing

Users, permissions & daily workflows

Pharmacy staff

Search and scan items
Create POS sales / record tender
View available stock
Dispense against prescription
Receive stock-count assignments

Storekeeper

Receive and transfer stock
Maintain central store balance
Prepare count sessions
Review discrepancy reasons

HOD / pharmacy lead

Manage catalogue and fixed prices
Approve stock adjustments
Manage pharmacy units and staff
Review sales, expiry and count reports

MD / management

View authorized reports remotely
Connect through hospital VPN
Review backup and service status

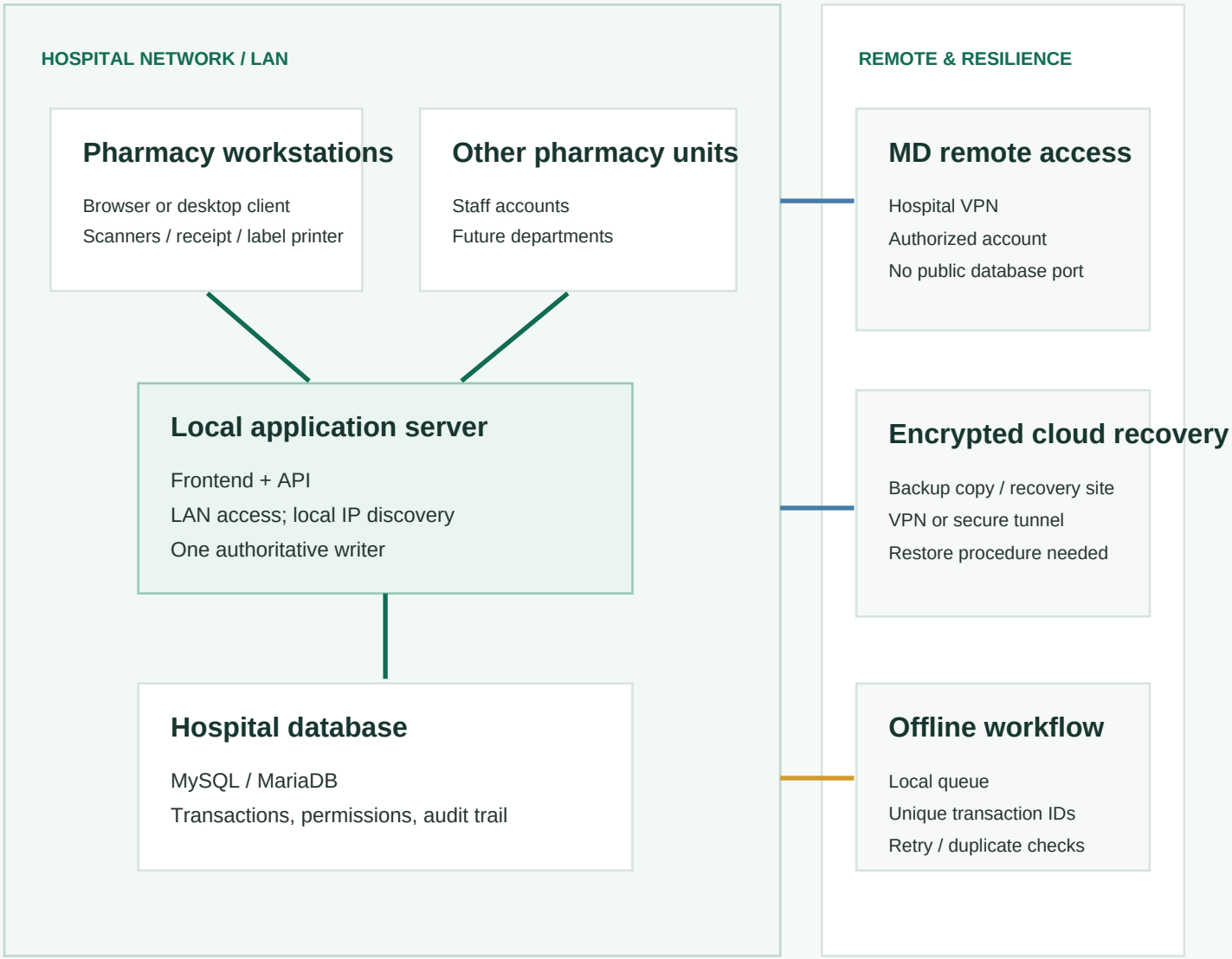
CORE TRANSACTION PATH

1 **Receive** — 2 **Store / transfer** — 3 **Sell / dispense** — 4 **Count** — 5 **Reconcile**

CONTROL RULES

Use named accounts and role-based permissions; log stock movements, adjustments and returns.
A stock adjustment records system quantity, counted quantity, variance, reason and approver.
Keep local sales available during network interruptions; sync safely when connectivity returns.

System architecture & deployment



KEY DEPLOYMENT RECOMMENDATION

KEEP THE HOSPITAL LAN FAST AND THE DATA CONSISTENT

Local server and local database serve pharmacy users during normal operation.

Use encrypted, scheduled backups to cloud storage or an approved recovery server.

Avoid multi-master writes until conflict handling and recovery are deliberately designed.

Define backup schedule, retention, recovery time and recovery point with hospital management.

Page design | dashboard & inventory

BKD PHARMACY

- Dashboard
- Inventory
- Point of Sale
- Stock Counts
- Patients
- Reports
- Staff Chat

Pharmacy overview

Today sales	Items low
245,600	08

Near expiry	Pending counts
12	02

Sales trend / stock alerts / recent activity

INVENTORY WORKSPACE

BKD PHARMACY

- Dashboard
- Inventory
- Point of Sale
- Stock Counts
- Patients
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- Staff Chat

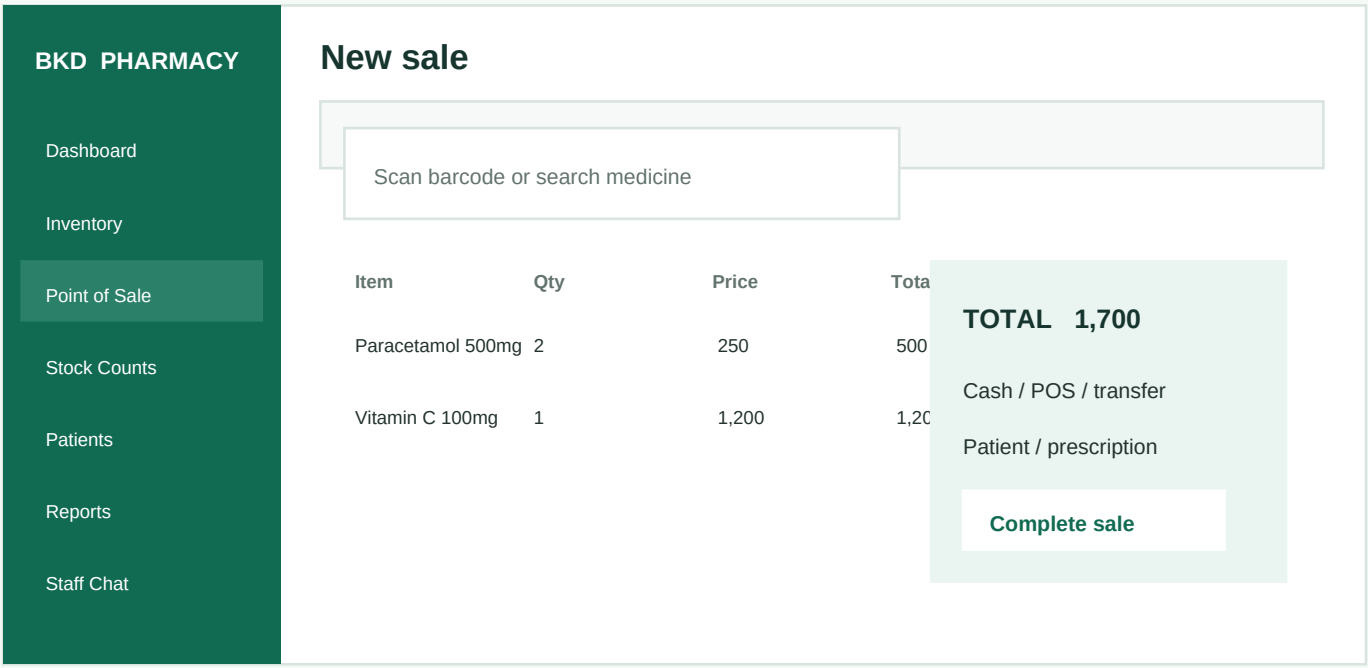
Stock balances

Search / Notifications / Profile

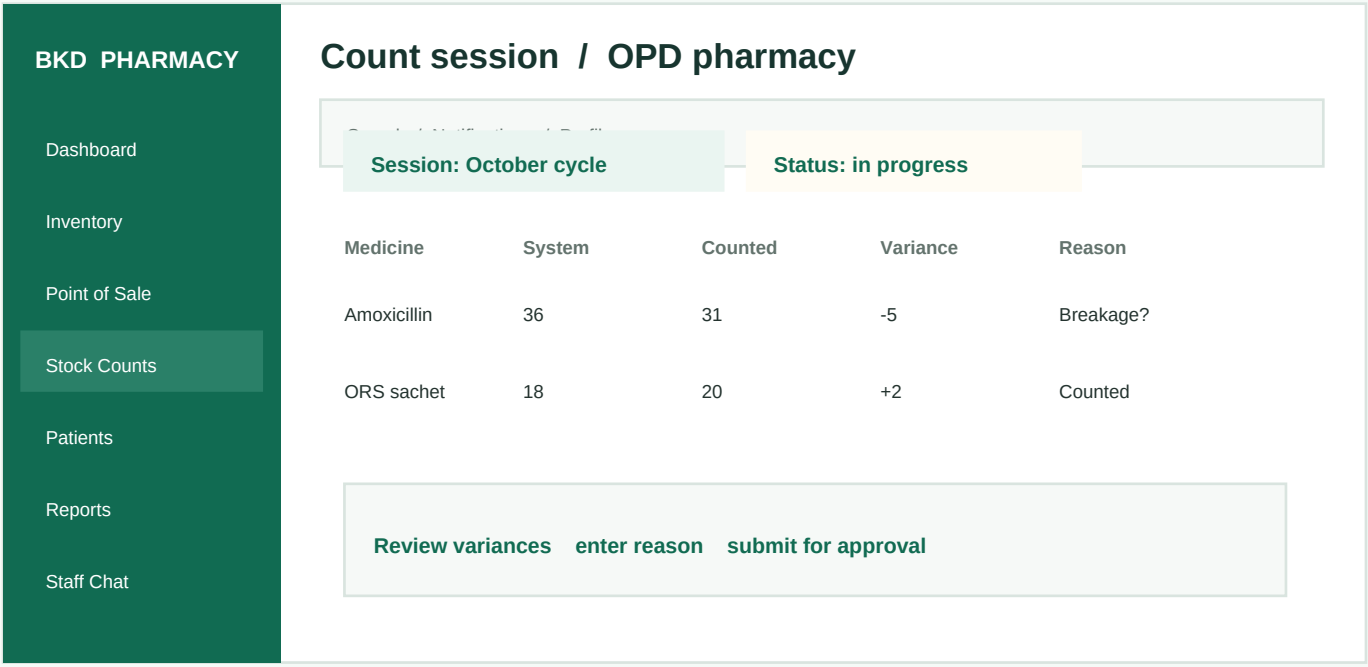
Medicine	Unit	On hand	Reorder	Expiry
Paracetamol 500mg	Central	420	100	2027-04
Amoxicillin 250mg	OPD	36	60	2026-12
ORS sachet	Main	18	50	2027-01

Page design | POS & physical stock count

POINT OF SALE



PHYSICAL STOCK COUNT



Page design | patients & reports

PATIENT AND PRESCRIPTION-LINKED DISPENSING

BKD PHARMACY

- Dashboard
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- Staff Chat

Patient record / prescriptions

Patient: A. Sample | Hospital number: GH-004281

Prescription: Dr. Example | 18 Oct 2026

Medicine	Dose	Duration	Qty	Label
Amoxicillin 250mg	1 capsule	5 days	15	Preview
Paracetamol 500mg	2 tablets	3 days	18	Preview

Check stock and pharmacist directions before dispense

REPORTS AND MANAGEMENT VIEW

BKD PHARMACY

- Dashboard
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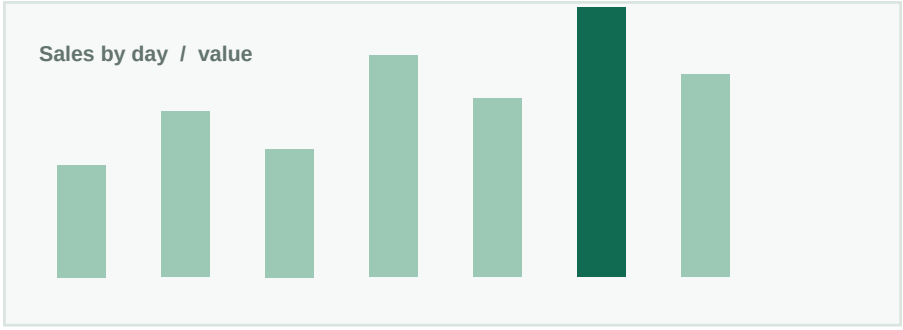
- Staff Chat

Sales and stock report

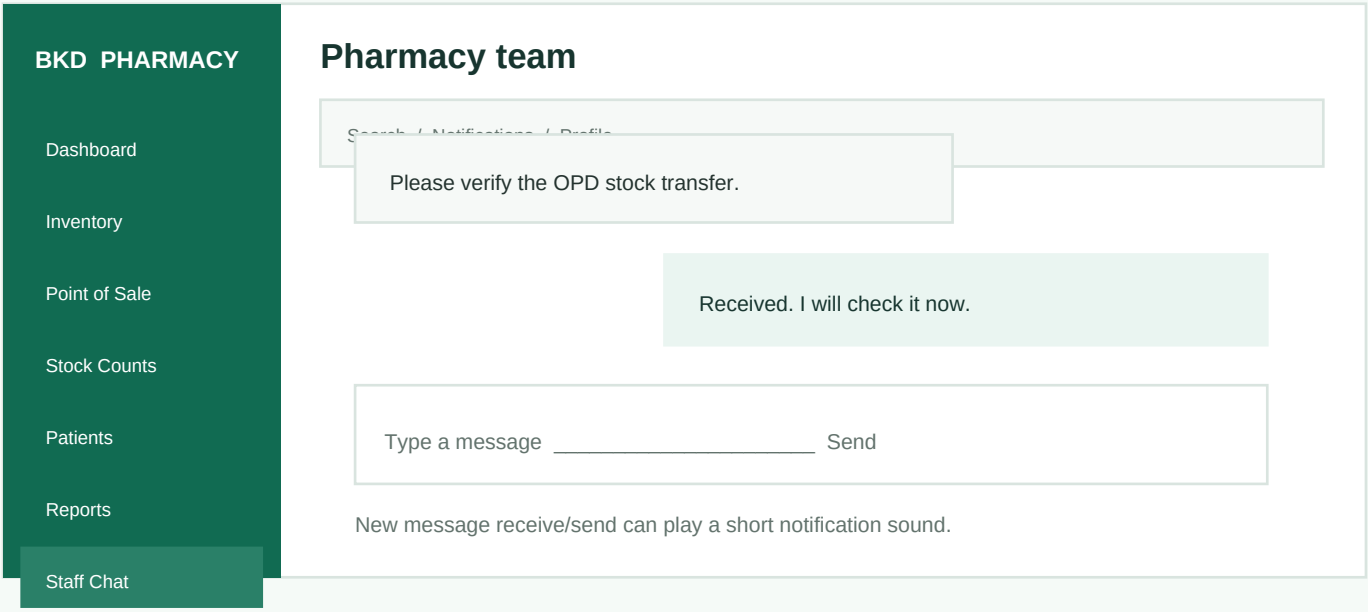
Daily | Weekly | Monthly



Export sales, stock movement, expiry and count variance reports by permission.

Page design | staff chat & account controls

STAFF CHAT (TEXT ONLY)



ACCOUNT & APPEARANCE

PROFILE SETTINGS

Name and contact information

Change password securely

Generated avatar (no media upload)

Notification and privacy settings

Account session controls

APPEARANCE CONTROLS

Hospital-green default theme

Choose sidebar and topbar colors

Adjust icons and interface accents

Store theme preference per account

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Privacy boundary

Staff chat supports text and built-in reactions. No image, video, document or voice-note uploads.

Implementation roadmap & open decisions

01 Foundation

Review requirements; prepare server and user roles; import item master and opening stock.

02 Stock operations

Receiving, transfers, batch/expiry tracking, physical counts and approved adjustments.

03 Sales & dispensing

POS, payment recording, receipt/label printing, patient and prescription workflows.

04 Resilience & reporting

Offline queue, synchronization, alerts, daily/weekly/monthly reporting and backups.

05 Hardening & rollout

Pilot pharmacy, staff training, restore drill, LAN access and VPN management access.

CONFIRM BEFORE IMPLEMENTATION SIGN-OFF

OPEN ITEMS

Count frequency, count freeze rules and approval thresholds
Payment-terminal model and integration method; mandatory app tender record
Server / cloud hosting preference, backup retention, RPO and RTO
Import file format and mapping; printers/scanners and medicine label layout
Client accepts NHIS later; confirm any controlled-drug legal requirements when scoped